Dvojvyberove testy, ANOVA

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```
library(latexpdf)
```

Dva nezavisle vybery

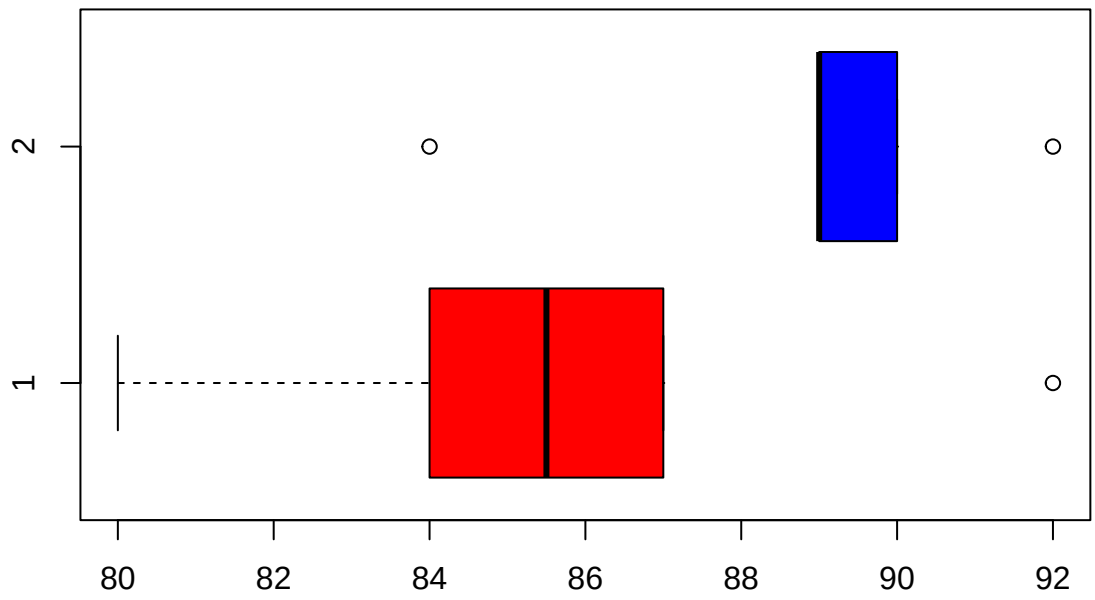
Najrv test o rovnosti, nerovnosti disperzii

Priklad Dva druhy plastickej hmoty (vyroba plastovych okien) sme vystavili svetlu po dobu 6 mesiacov. Odolnost voci svetlu v dnoch sme zaznamenali. Vyroba prveho plastu je lacnejsia. Zistite, ci su rozdiely v odolnosti voci svetlu pre dva druhy plastov $\alpha = 0.05$.

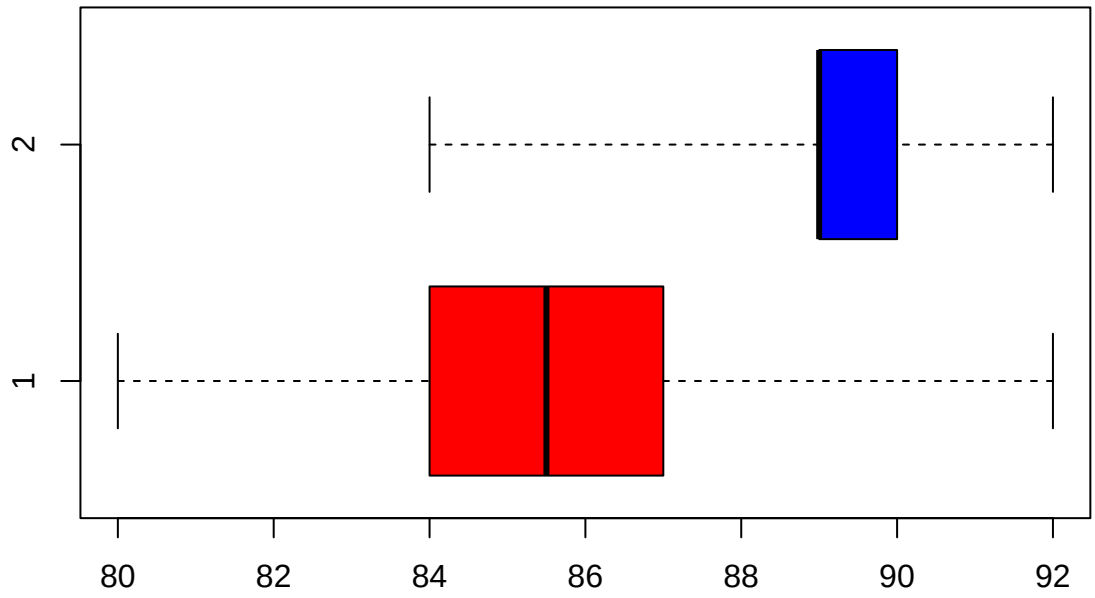
```
prvy <- c(85,87,92,80,84,86)
druhy <- c(89,89,90,84,92)
```

Krabicove grafy

```
boxplot(prvy,druhy,col=c("red","blue"),horizontal = T)
```




```
boxplot(prvy,druhy,col=c("red","blue"),horizontal = T,range=0)
```




```
shapiro.test(prvy)#su normalne rozdelene
```

```
##
##  Shapiro-Wilk normality test
##
## data:  prvy
## W = 0.96665, p-value = 0.8692
```

```
shapiro.test(druhy)# su normalne rozdelene
```

```
##
##  Shapiro-Wilk normality test
##
## data:  druhy
## W = 0.88783, p-value = 0.3463
```

```
var.test(prvy,druhy)
```

```
##
##  F test to compare two variances
##
## data:  prvy and druhy
## F = 1.7778, num df = 5, denom df = 4, p-value = 0.5971
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##   0.1898428 13.1340191
## sample estimates:
## ratio of variances
```

```
##           1.777778
```

Phodnota >0.05, nezamietam H_0. Disperzie sa rovnaju.

Test strednych hodnot

```
t.test(prvy,druhy,paired = F,var.equal = T)
```

```
##
##  Two Sample t-test
##
## data:  prvy and druhy
## t = -1.466, df = 9, p-value = 0.1767
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -7.968424  1.701757
## sample estimates:
## mean of x mean of y
##  85.66667  88.80000
```

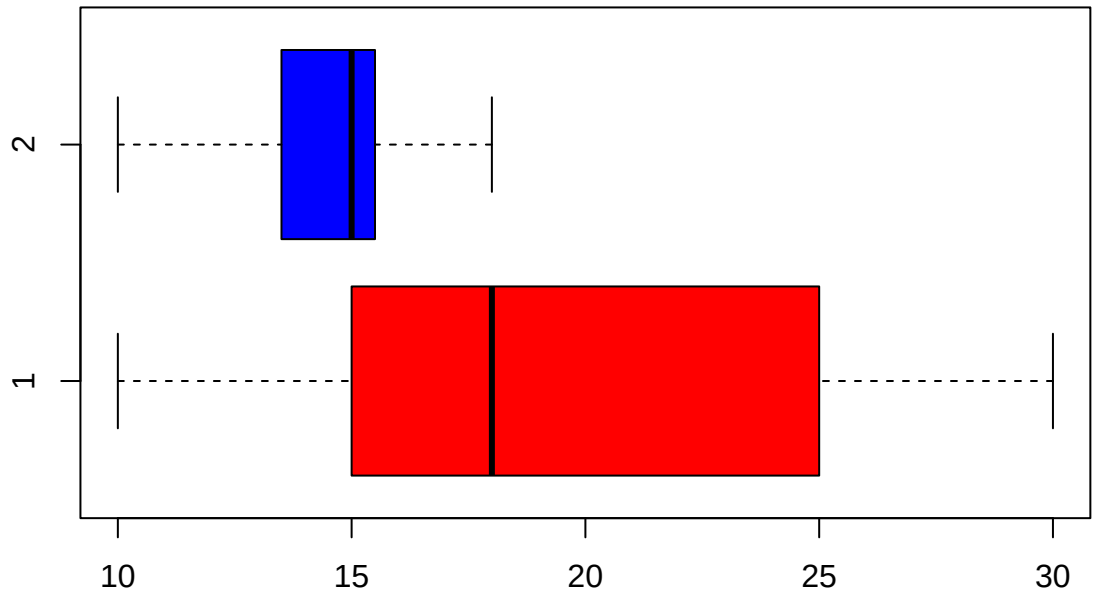
Phodnota>0.05, nezamietam hypotezu o rovnosti strednych hodnot. Default pre t.test var.equal=F

```
#t.test(prvy,druhy,paired = F,var.equal = F)
```

Dvojvyberovy, nerovnake disperzie

Priklad Firma odobera ochranne pracovne pomocky od dvoch dodavatelov AA, BB. Zaznamenali sme tieto dlzky dodavok od objednavky. Na hladine vyznamnosti $\alpha = 0.05$ testujte hypotezu, ze dlzky dodavok su rovnake.

```
aa <- c(10,12,15,25,18,20,15,25,30)
bb <- c(15,15,18,10,16,12,15)
boxplot(aa,bb,col=c("red","blue"),horizontal = T,range=0)
```




```
shapiro.test(aa)#ok
```

```
##
##  Shapiro-Wilk normality test
##
## data:  aa
## W = 0.95341, p-value = 0.7274
```

```
shapiro.test(bb)#ok
```

```
##
##  Shapiro-Wilk normality test
##
## data:  bb
## W = 0.9206, p-value = 0.474
```

```
var.test(aa,bb)#test pre rovnost disperzii
```

```
##
##  F test to compare two variances
##
## data:  aa and bb
## F = 6.4167, num df = 8, denom df = 6, p-value = 0.03593
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##   1.14591 29.84838
## sample estimates:
## ratio of variances
```

```
##           6.416667
```

Zamietame hypotezu o rovnosti disperzii

```
t.test(aa,bb,paired = F,var.equal = F)
```

```
##
##  Welch Two Sample t-test
##
## data:  aa and bb
## t = 1.8286, df = 10.941, p-value = 0.09483
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.911966  9.832601
## sample estimates:
## mean of x mean of y
##  18.88889  14.42857
```

Phodnota>0.05, nezamietame hypotezu o rovnosti strednych hodnot.

ANOVA

Test rovnosti strednych hodnot pre viac ako dve skupiny

Je nutne aby, boli splnene dve podmienky

normalita dat v triedach

rovnosti disperzi v triedach

Priklad: Nase vyukove data. K dispozicii mame dat o priemernom prijme pre tri podmnoziny (faktor vzdelanie-zs, ss, vs). Testume hypotezu o rovnosti strednych hodnot, ze faktor druh vzdelania je statisticky nevyznamny.

```
library(readxl)
library(RColorBrewer)
library(vioplot)
```

```
## Loading required package: sm
```

```
## Package 'sm', version 2.2-6.0: type help(sm) for summary information
```

```
## Loading required package: zoo
```

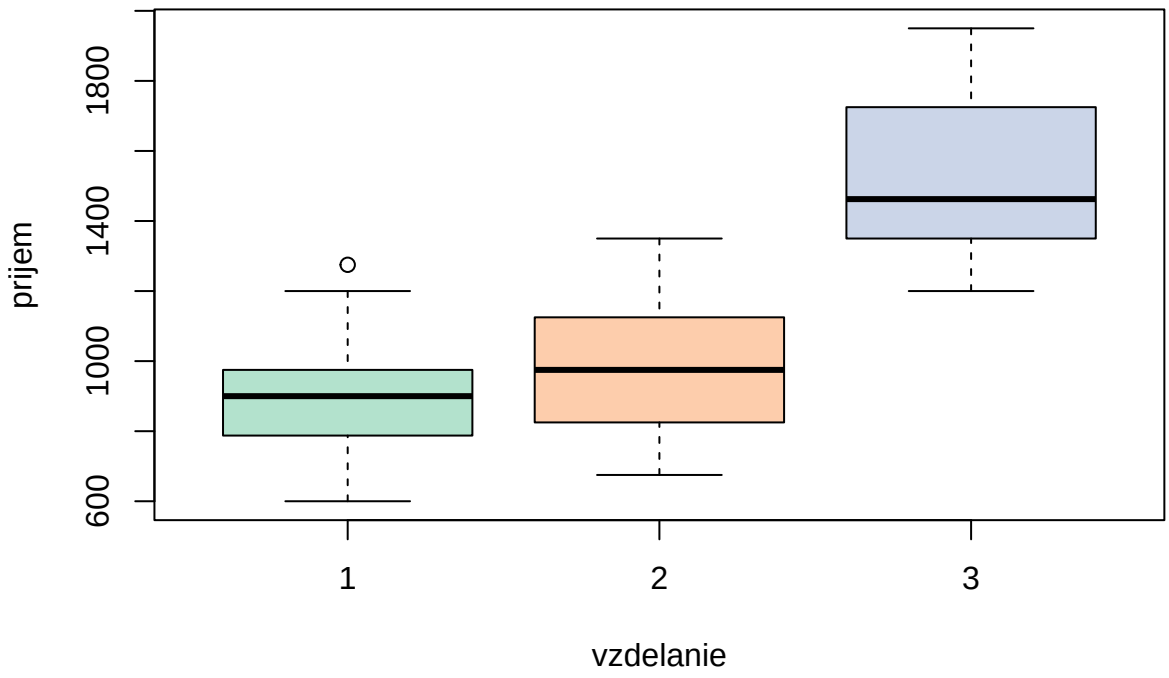
```
##
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

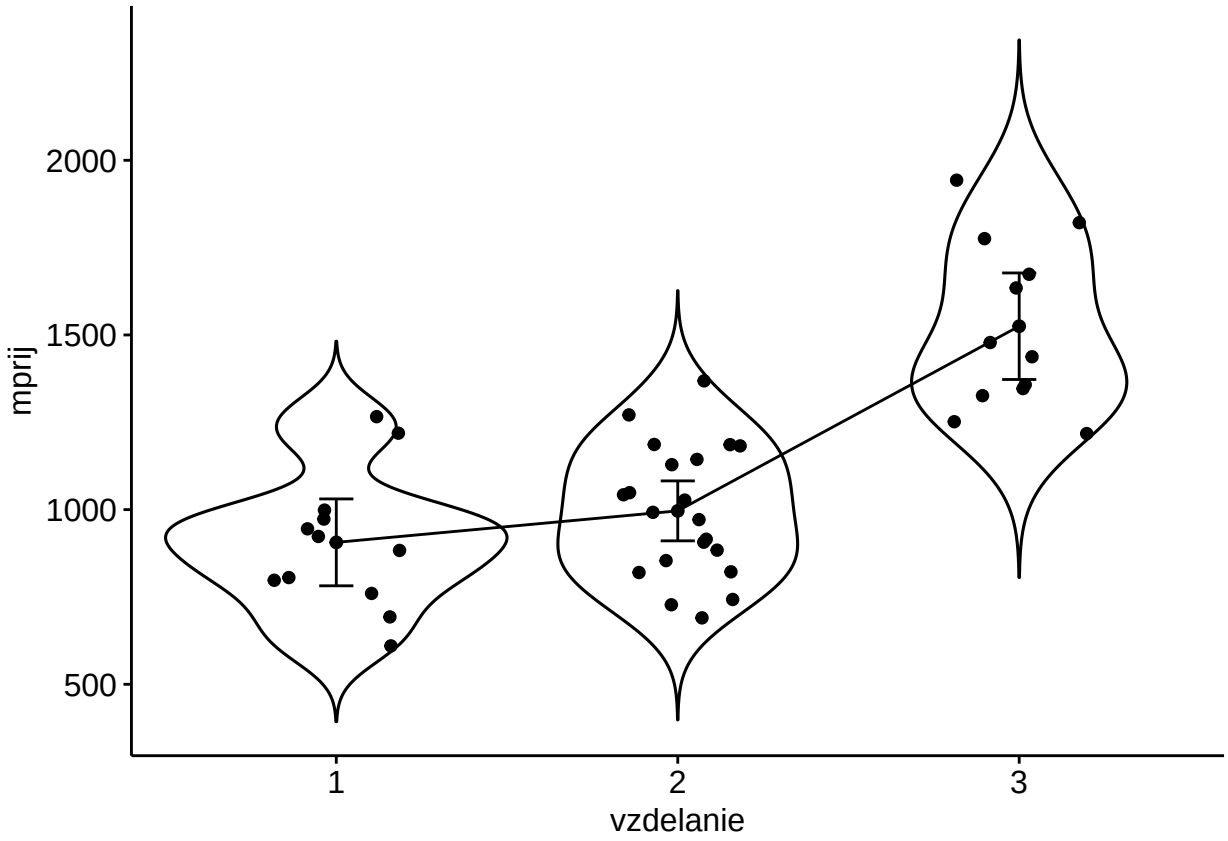
```
library(ggplot2)
library(ggpubr)
data <- read_xlsx("data_vyuka.xlsx")
new_data <- na.omit(data)
```

Najprv grafy

```
boxplot(new_data$mprij~new_data$vzdelanie,col=brewer.pal(3,"Pastel2"),
        xlab="vzdelanie",ylab="prijem")
```




```
ggline(new_data,x="vzdelanie",y="mprij",add=c("mean_ci","violin",
                                              "jitter"))
```



Overenie podmienok pre ANOVA normalita dat

```
tapply(new_data$mprij, new_data$vzdelanie, shapiro.test)
```

```
## $`1`
##
##  Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.9556, p-value = 0.7197
##
##
## $`2`
##
##  Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.96792, p-value = 0.6867
##
##
## $`3`
##
##  Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.93161, p-value = 0.3975
```

Data v triedach su normalne rozdelene.

Rovnost disperzii

```
bartlett.test(new_data$mprij, new_data$vzdelanie)
```

```
##
##  Bartlett test of homogeneity of variances
##
## data:  new_data$mprij and new_data$vzdelanie
## Bartlett's K-squared = 0.89242, df = 2, p-value = 0.6401
```

Phodnota>0.05, disperzie v triedach sa rovnaju, ANOVA

Faktorizujeme

```
vzdelanie <- factor(new_data$vzdelanie)
vzdelanie
```

```
##  [1] 3 2 1 2 3 1 3 1 2 2 2 2 2 3 2 1 3 3 2 1 2 2 2 3 2 2 3 3 1 2 1 2 2 1 3 3 1 2
## [39] 2 3 2 2 1 1 1
## Levels: 1 2 3
```

```
View(new_data)
anova <- aov(new_data$mprij~vzdelanie)
anova
```

```
## Call:
##    aov(formula = new_data$mprij ~ vzdelanie)
##
## Terms:
##                 vzdelanie Residuals
## Sum of Squares    2835237   1765513
## Deg. of Freedom         2        42
##
## Residual standard error: 205.0269
## Estimated effects may be unbalanced
```

```
summary.aov(anova)
```

```
##             Df  Sum Sq Mean Sq F value   Pr(>F)
## vzdelanie    2 2835237 1417618   33.72 1.84e-09 ***
## Residuals   42 1765513   42036
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Vystupom je standardna ANOVA tabulka, Pr(>F)-Probalility Value< 0.05, teda zamietame hypotezu o rovnosti strednych hodnot. Faktor vzdelanie je nenulovy, statisticky vyznamny. To jest vzdelanie ma vplyv na vysku prijmu.

Nasleduju teda post hoc testy, kde zistime, ktore dvojice su odlisne a vypocet miery efektu (ci rozdiel je silny, ma prakticky vyznam)

Miera efektu

```
library(effectsize)
eta_squared(anova)
```

```
## For one-way between subjects designs, partial eta squared is equivalent
##   to eta squared. Returning eta squared.
```

```
## # Effect Size for ANOVA
##
## Parameter | Eta2 |       95% CI
## ------------------------------
## vzdelanie | 0.62 | [0.45, 1.00]
##
## - One-sided CIs: upper bound fixed at [1.00].
```

Cislo je vysoke, rozdiely su aj prakticky vyznamne.

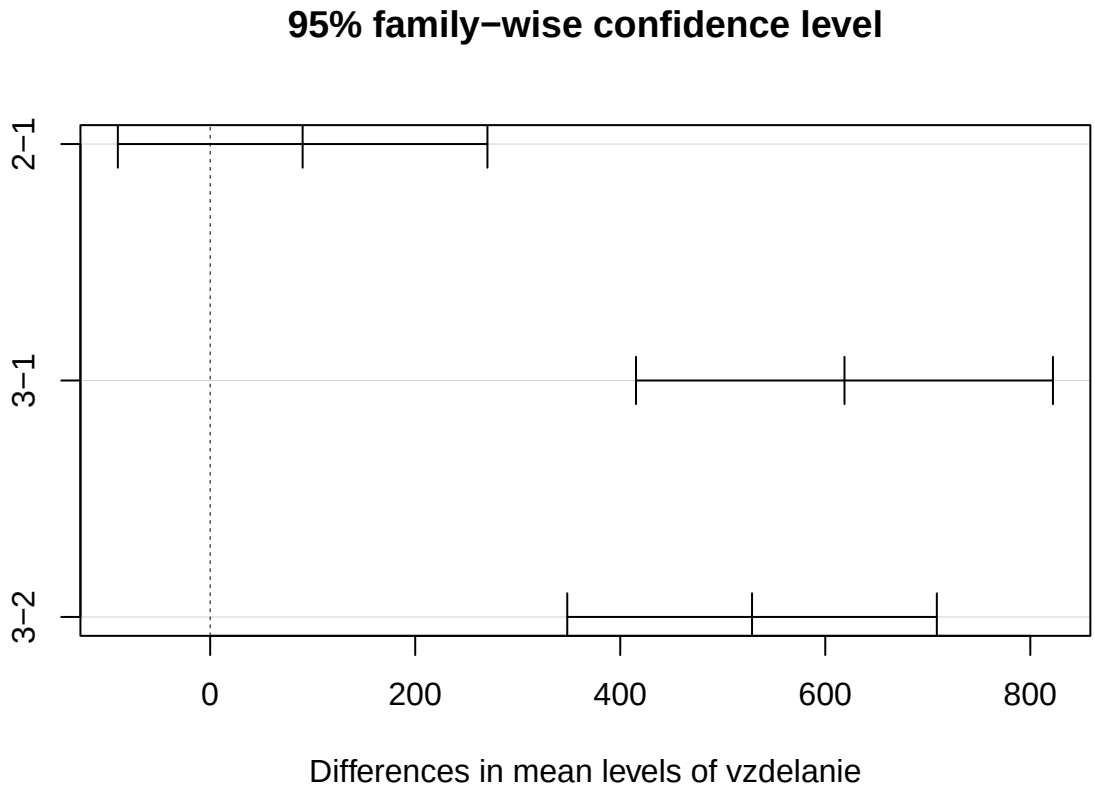
Post hoc test, H_0 je rovnost strednych hodnot, testujeme po dvojiciach.

```
TukeyHSD(anova)
```

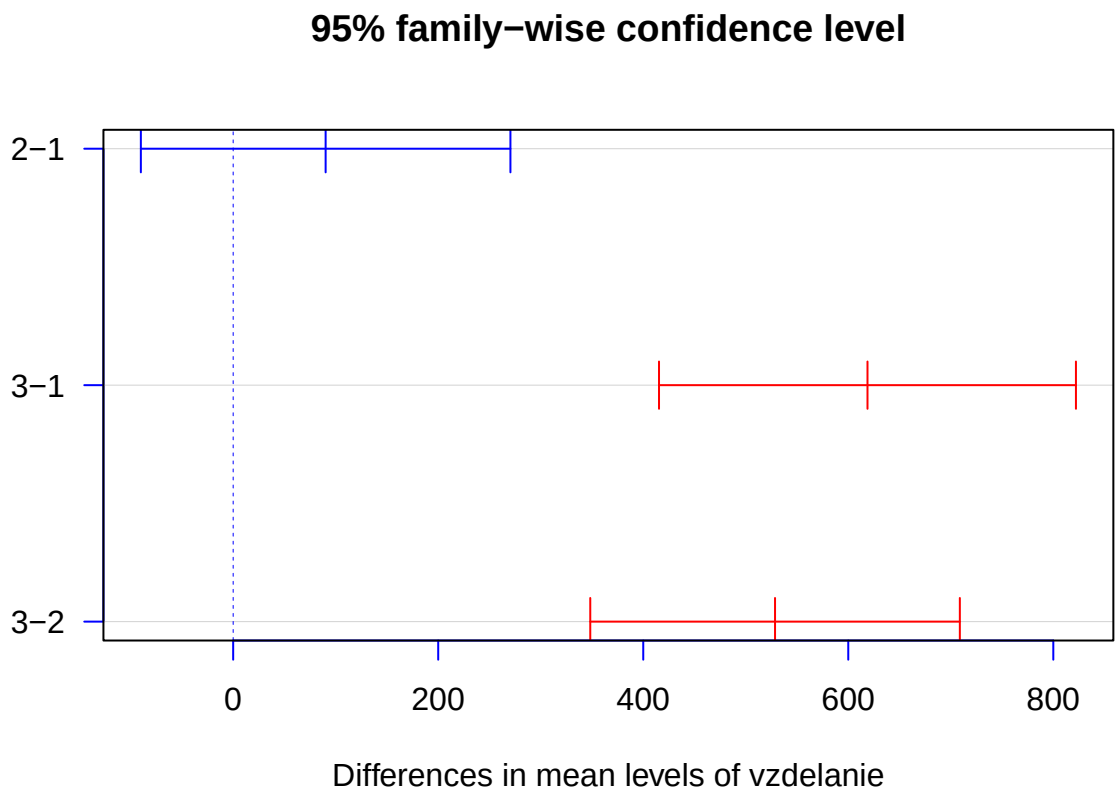
```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = new_data$mprij ~ vzdelanie)
##
## $vzdelanie
##          diff       lwr      upr     p adj
## 2-1  90.17857 -90.07478 270.4319 0.4508302
## 3-1 618.75000 415.39679 822.1032 0.0000000
## 3-2 528.57143 348.31808 708.8248 0.0000000
```

Triedy 1-2, nezamietame H_0 rovnake stredne hodnoty. Triedy 1-3, 2-3 rozdielne stredne hodnoty (podla Padjust<0.05). Teraz IS, ktory obsahuje nulu->triedy nie su odlisne.

```
plot(TukeyHSD(anova))
```



```
plot(TukeyHSD(anova),las=1,col=c("blue","red","red"))
```



Iny post test je Scheffe post test, DescTools

```
library(DescTools)
ScheffeTest(anova)
```

```
##
##   Posthoc multiple comparisons of means: Scheffe Test
##     95% family-wise confidence level
##
## $vzdelanie
##          diff    lwr.ci   upr.ci    pval
## 2-1  90.17857 -98.10248 278.4596  0.4839
## 3-1 618.75000 406.34033 831.1597 2.5e-08 ***
## 3-2 528.57143 340.29038 716.8525 5.9e-08 ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

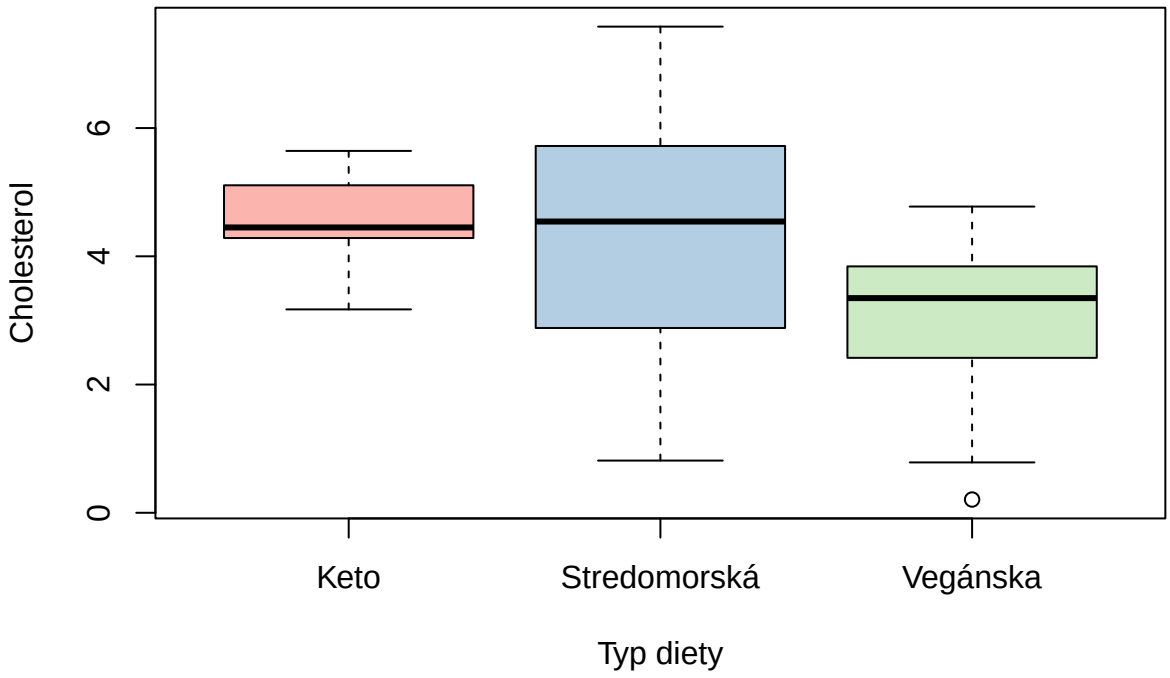
Porusenie predpokladu rovnosti rozptylov

Welch ANOVA-data chlesterol

```
data <- read_xlsx("cholesterol.xlsx")
View(data)
```

Faktor-druh diety. Otazka - ma faktor druh diety vplyv na vysku cholesterolu?

```
boxplot(data$Cholesterol~data$Dieta,col=brewer.pal(3,"Pastel1"),
        xlab="Typ diety",ylab="Cholesterol")
```



normalita

```
tapply(data$Cholesterol, data$Dieta, shapiro.test)
```

```
## $Keto
##
##  Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.96209, p-value = 0.35
##
##
## $Stredomorská
##
##  Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.9579, p-value = 0.2735
##
##
## $Vegánska
##
##  Shapiro-Wilk normality test
```

```
##
## data:  X[[i]]
## W = 0.93428, p-value = 0.06386
```

Normalita ok.

Rovnost rozptylov

```
bartlett.test(data$Cholesterol,data$Dieta)
```

```
##
##  Bartlett test of homogeneity of variances
##
## data:  data$Cholesterol and data$Dieta
## Bartlett's K-squared = 34.699, df = 2, p-value = 2.919e-08
```

Nie je splnena, Welch ANOVA.

Faktorizujeme.

```
Dieta <- factor(data$Dieta)
anova <- oneway.test(data$Cholesterol~data$Dieta,var.equal = F)
print(anova)# nie v tabulke ako sme zvyknuti, ale
```

```
##
##  One-way analysis of means (not assuming equal variances)
##
## data:  data$Cholesterol and data$Dieta
## F = 21.036, num df = 2.000, denom df = 50.412, p-value = 2.278e-07
```

Phodnota je

```
library(rstatix)#kvoli nasledujucemu testu, post hoc ak nerovnake disp.
```

```
##
## Attaching package: 'rstatix'
```

```
## The following objects are masked from 'package:effectsize':
##
##     cohens_d, eta_squared, omega_squared
```

```
## The following object is masked from 'package:stats':
##
##     filter
```

```
games_howell_test(Cholesterol ~ Dieta,data=data)
```

```
## # A tibble: 3 x 8
##   .y.         group1     group2 estimate conf.low conf.high   p.adj p.adj.signif
## * <chr>       <chr>      <chr>     <dbl>    <dbl>     <dbl>   <dbl> <chr>
## 1 Cholesterol Keto       Stred~   -0.257    -1.18     0.661 7.73e-1 ns
## 2 Cholesterol Keto       Vegán~   -1.46     -2.00    -0.916 1.26e-7 ****
## 3 Cholesterol Stredomor~ Vegán~   -1.20     -2.18    -0.215 1.36e-2 *
```

Nie je rozdiel keto-stredomorska, je rozdiel keto-vega, stredomorska-vegan.

```
################################################
```

Dvojfaktorova analyza rozptylu

Data sa rozdeluju do skupin podla dvoch faktorov, riadkovy, stlpcovy faktor. Testuju sa vlastne dve hypotezy.

Vplyv riadkoveho faktora α_i je nula,

Vplyv stlpcoveho faktora β_j je nula,

Zatial bez interakcii, kazda z tried ij musi splnat podmienky normality a triedy maju rovnake disperzie.

Traja laboranti (faktor laborant) urobili opakovane merania deg na styroch chromatografoch (faktor pristroj). Predpokladajme, ze data v triedach su normalne rozdelene a s rovnakou disperziou vizualizujte, testujte hypotezy dvofaktorovej ANOVA.

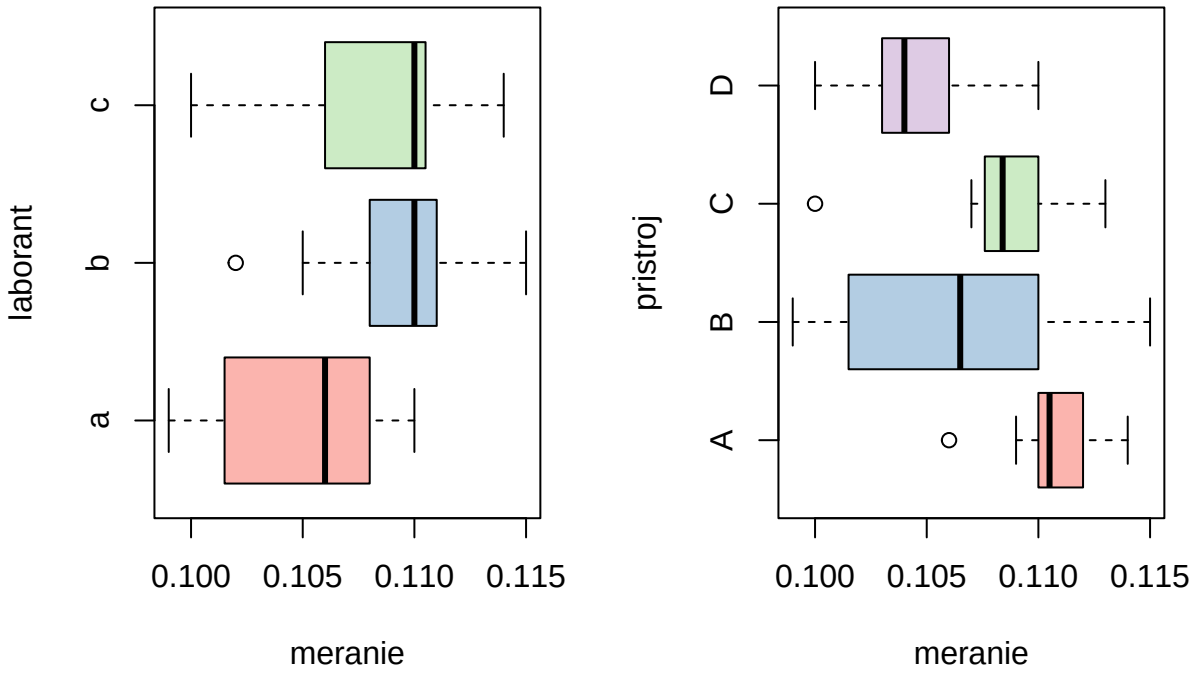
```
anova1 <- readxl::read_xlsx("anova1.xlsx")
View(anova1)
```

Najprv faktorizujeme.

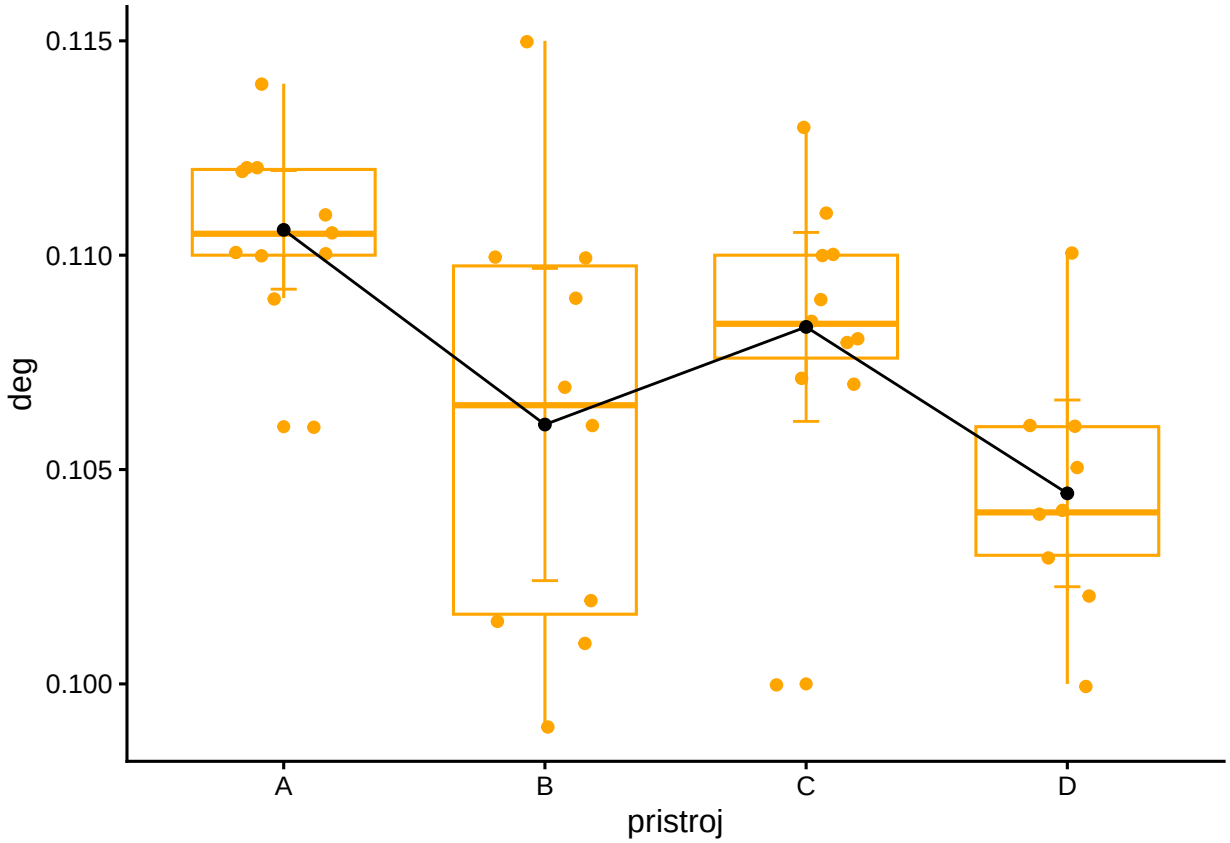
```
laborant <- factor(anova1$laborant)
laborant
```

```
##  [1] a a a a a a a a a a a a a a a b b b b b b b b b b b b b b b c c c c c c c c c c c
## [39] c c c
## Levels: a b c
```

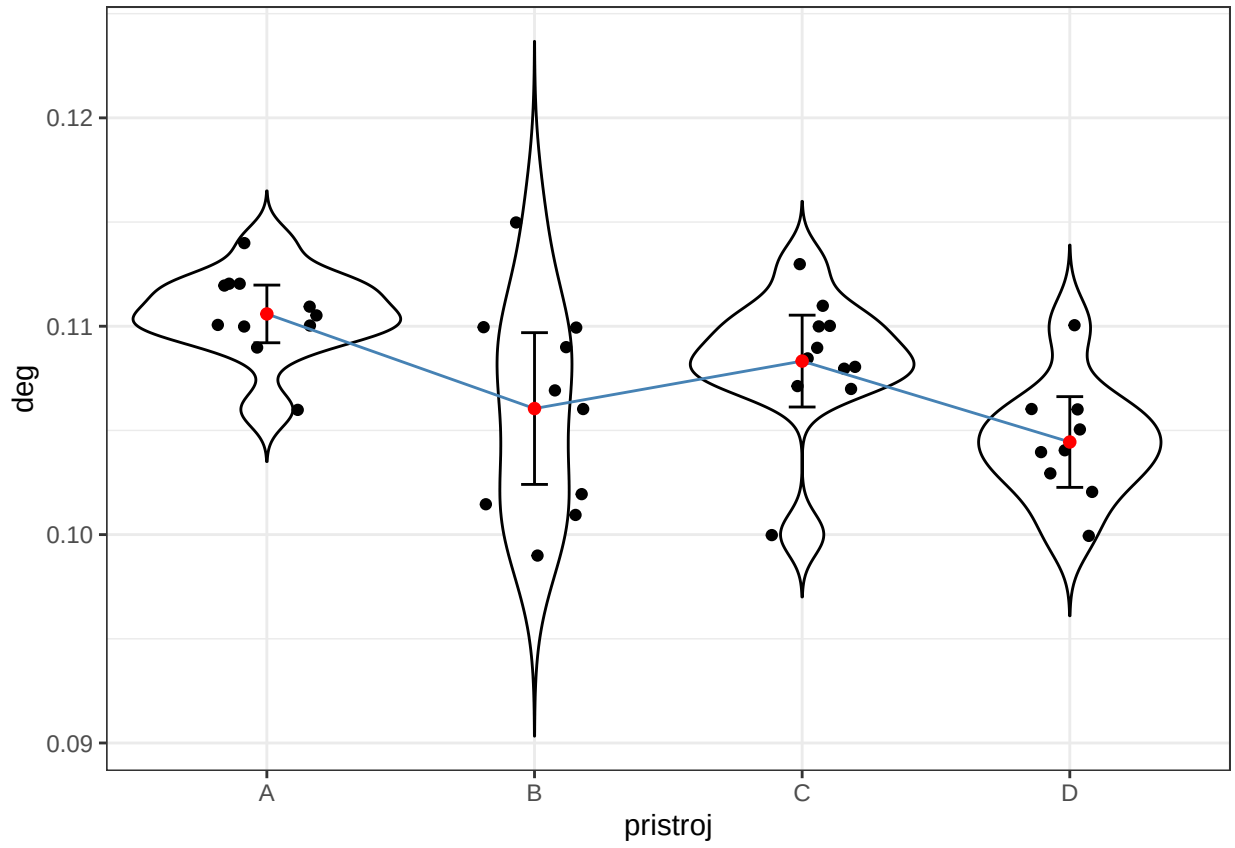
```
pristroj <- factor(anova1$pristroj)
par(mfrow=c(1,2))
boxplot(anova1$deg~anova1$laborant,col=brewer.pal(3,"Pastel1"),
        xlab="meranie",ylab= "laborant",horizontal=T)
boxplot(anova1$deg~anova1$pristroj,col=brewer.pal(4,"Pastel1"),
        xlab="meranie",ylab="pristroj",horizontal=T)
```



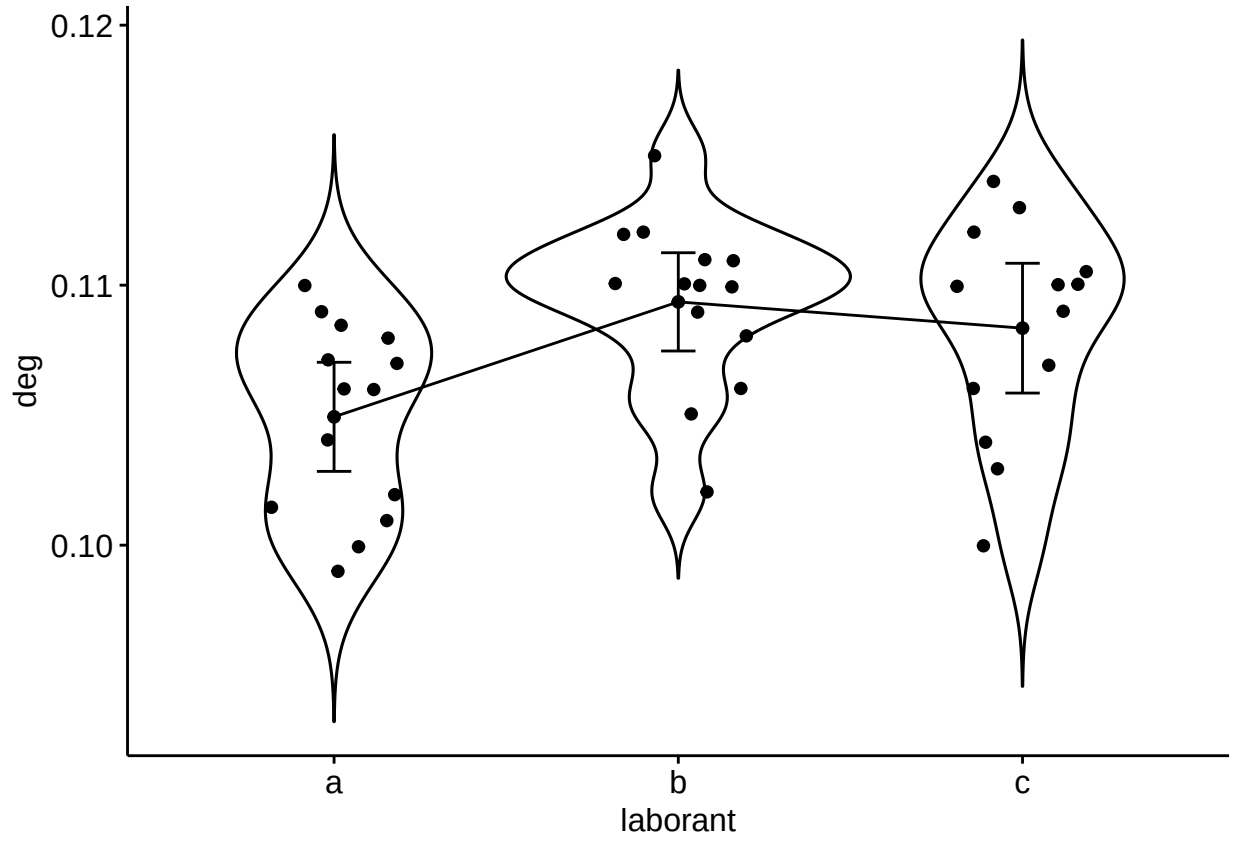
```
ggline(anova1,x="pristroj",y="deg",col ="orange",
       add=c("mean_ci","boxplot","jitter"),color="black")+
  theme_classic2()
```




```
ggline(anova1,x="pristroj",y="deg",col ="black",
       add=c("mean_ci","violin","jitter"),point.color="red",
       color="steelblue")+theme_bw()
```




```
ggline(anova1,x="laborant",y="deg",
       add = c("mean_ci","violin","jitter"))
```



Testy normality a rovnosti disperzie

```
tapply(anova1$deg, list(anova1$laborant, anova1$pristroj),
       function(x) shapiro.test(x)$p.value )
```

```
##           A         B         C         D
## a 0.4632629 0.3689993 0.5130401 0.6368868
## b 0.2724532 0.8776501 0.9718771 0.7262263
## c 0.5835206 0.6368868 0.4243509 0.6368868
```

```
levene_test(deg~laborant*pristroj,data=anova1)
```

```
## # A tibble: 1 x 4
##     df1   df2 statistic     p
##   <int> <int>     <dbl> <dbl>
## 1    11    29      1.16 0.353
```

Samotna ANOVA, tvarime sa ze

```
an1 <- aov(anova1$deg~laborant+pristroj)
summary(an1)
```

```
##             Df    Sum Sq   Mean Sq F value   Pr(>F)
## laborant     2 0.0001496 7.481e-05   8.227 0.001179 **
## pristroj     3 0.0001972 6.574e-05   7.229 0.000672 ***
## Residuals   35 0.0003183 9.090e-06
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Laborant: Phodnota<0.05, zamietam hypotezu o nulovosti faktora- laborant, laborant ma vplyv na vysledky merani.

Pristroj: Phodnota<0.05, zamietam hypotezu o nulovosti faktora - pristroj, pristroj ma vplyv na vysledky merani.

Opat effect size, obidve hodnoty hovoria o velkej miere efektu faktorov.

```
eta_squared(an1)
```

```
##  laborant  pristroj
## 0.2249544 0.2965104
```

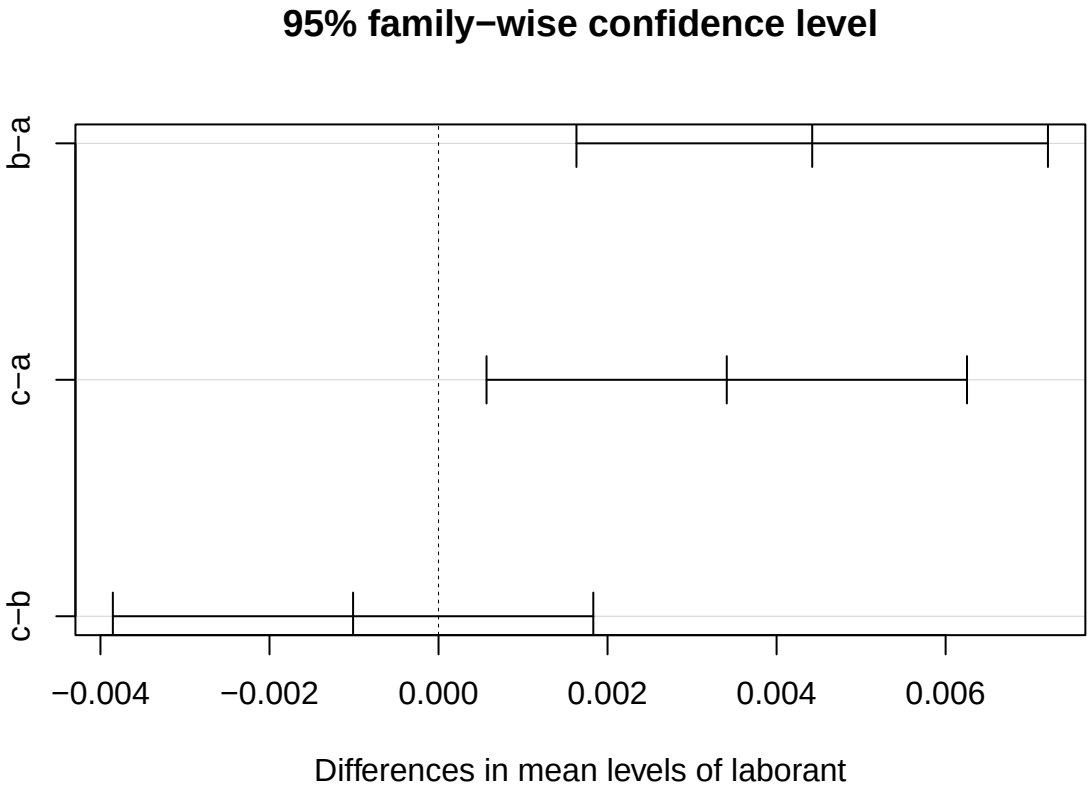
Post testy.

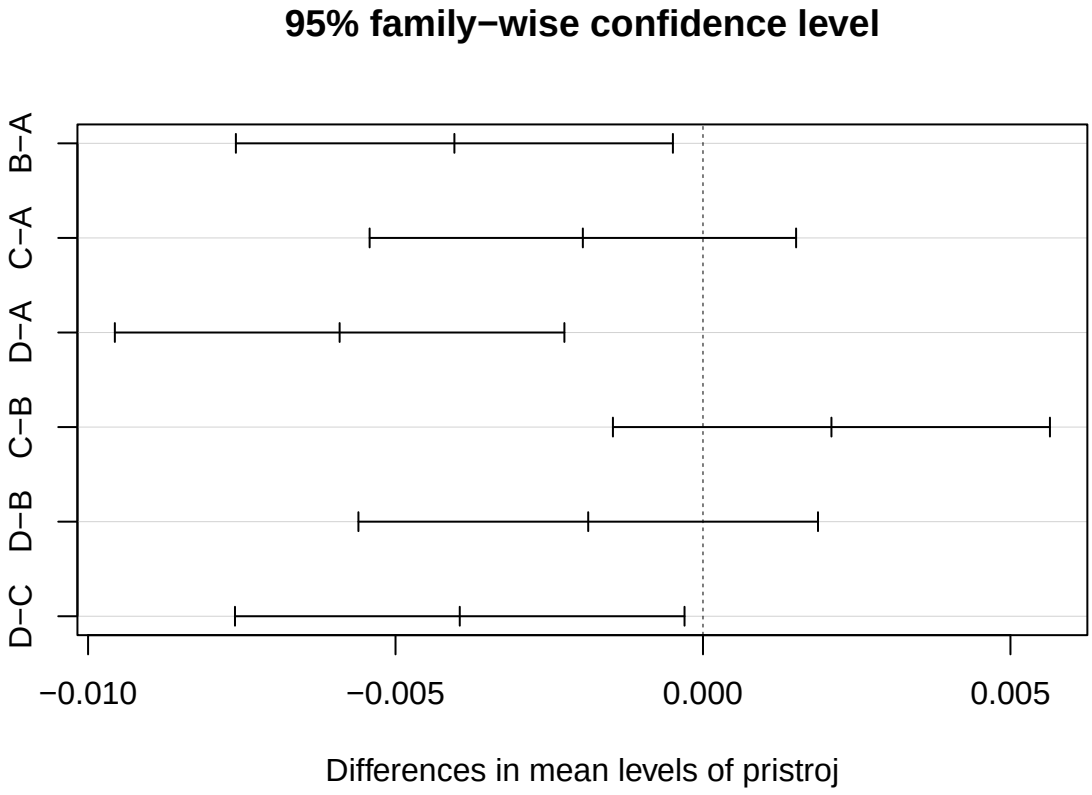
```
TukeyHSD(an1)#zakladny
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = anova1$deg ~ laborant + pristroj)
##
## $laborant
##             diff           lwr         upr     p adj
## b-a  0.004421429  0.0016320534 0.007210804 0.0012560
## c-a  0.003410440  0.0005679287 0.006252950 0.0157068
## c-b -0.001010989 -0.0038534999 0.001831522 0.6622408
##
## $pristroj
##             diff          lwr           upr     p adj
## B-A -0.004042517 -0.007595980 -0.0004890554 0.0206029
## C-A -0.001953596 -0.005421420  0.0015142275 0.4371571
## D-A -0.005909135 -0.009564543 -0.0022537280 0.0006058
## C-B  0.002088921 -0.001464541  0.0056423832 0.3998229
## D-B -0.001866618 -0.005603367  0.0018701312 0.5400054
## D-C -0.003955539 -0.007610946 -0.0003001316 0.0297302
```

Nenulove rozdiely su laborant a-b,a-c a pristroj A,B aA,D a C,D.

```
plot(TukeyHSD(an1))
```


95% family−wise confidence level
b−a
c−a
c−b
−0.004
−0.002
0.000
0.002
0.004
0.006
Differences in mean levels of laborant



Uvazujme interakcie faktorov, testujeme este jednu hypotezu, ze interakcie su nulove.

```
an2 <- aov(anova1$deg~laborant+pristroj+laborant*pristroj)
summary(an2)
```

```
##                   Df    Sum Sq   Mean Sq F value   Pr(>F)
## laborant           2 1.496e-04 7.481e-05   9.574 0.000642 ***
## pristroj           3 1.972e-04 6.574e-05   8.413 0.000355 ***
## laborant:pristroj  6 9.168e-05 1.528e-05   1.956 0.105173
## Residuals         29 2.266e-04 7.810e-06
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
eta_squared(an2)
```

```
##          laborant          pristroj laborant:pristroj
##         0.2249544         0.2965104         0.1378425
```

P hodnota pre interakcie>0.05, teda interakcie nemaju vplyv na vysledky merani, vplyv je nulovy a miera efektu mala.

Interakcie

Testovanie rýchlosti a použiteľnosti E-shopu Ulohou je zistiť, ako sa spracovanie užívateľskej požiadavky cas mení v závislosti od dvoch kľúčových faktorov: Faktor A: Latencia siete 1. Nízka (bežné kancelárske pripojenie, Low Latency)

2. Vysoká (mobilné pripojenie v špičke, High Latency)

Faktor B: Zložitosť UI 1. Jednoduché UI (intuitívne, moderné rozhranie) 2. Zložité UI (zastaralé, preplnené rozhranie)

```
anova2 <- readxl::read_xlsx("anova_interakcie.xlsx")
head(anova2)
```

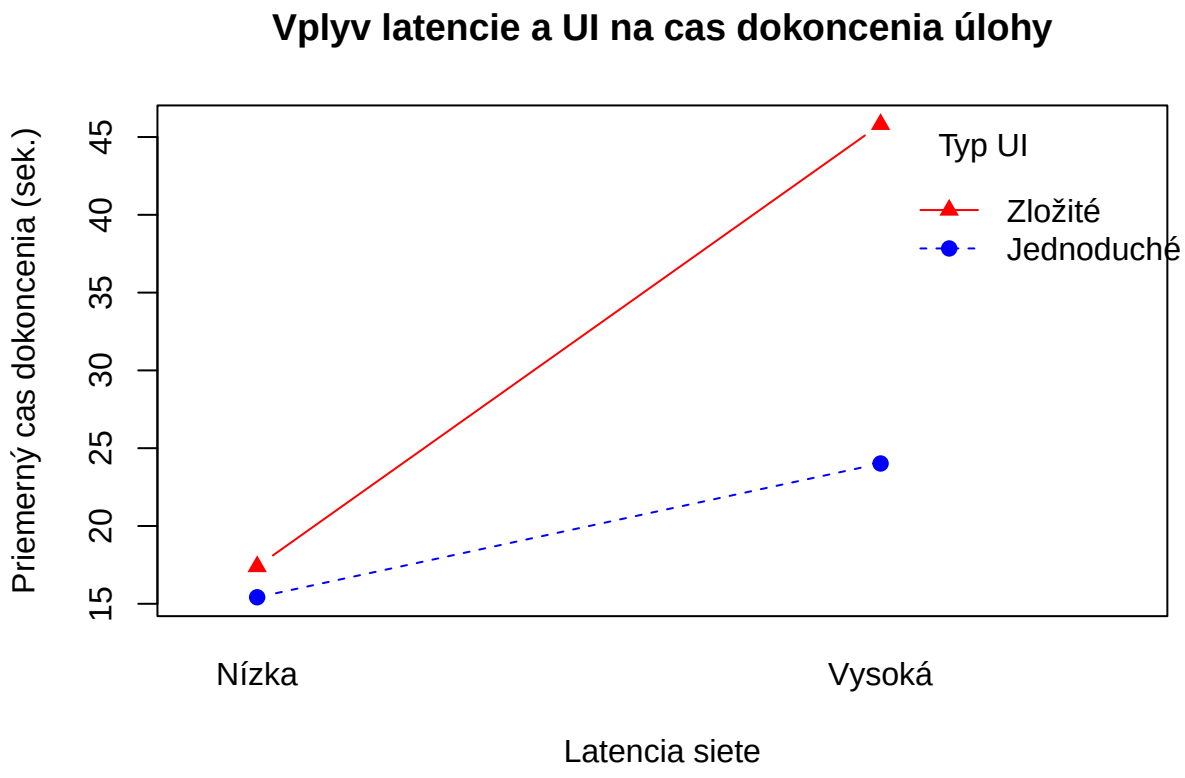
```
## # A tibble: 6 x 3
##     cas latencia ui
##   <dbl> <chr>    <chr>
## 1  16.1 Nízka    Jednoduché
## 2  14.0 Nízka    Jednoduché
## 3  15.0 Nízka    Jednoduché
## 4  17.2 Nízka    Jednoduché
## 5  17.9 Nízka    Jednoduché
## 6  12.8 Nízka    Jednoduché
```

```
an3 <- aov(cas ~ latencia * ui, data = anova2)
summary(an3)
```

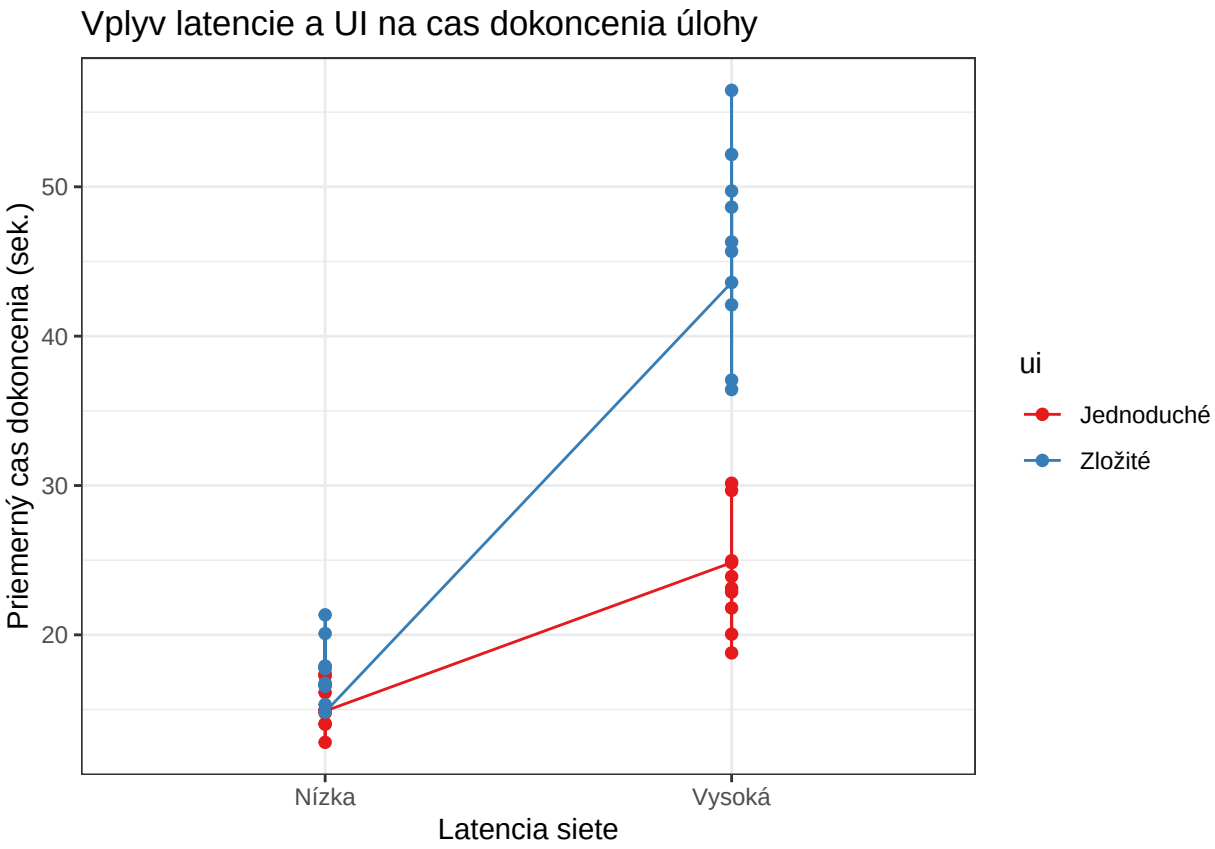
```
##             Df Sum Sq Mean Sq F value   Pr(>F)
## latencia     1   3426    3426  226.94  < 2e-16 ***
## ui           1   1413    1413   93.58 1.50e-11 ***
## latencia:ui  1    983     983   65.09 1.38e-09 ***
## Residuals   36    543      15
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Kedze sa interakcie potvrdili, tak teraz iba graf

```
par(mfrow=c(1,1))
interaction.plot(
  x.factor = anova2$latencia,
  trace.factor = anova2$ui,
  response = anova2$cas,
  fun = mean,
  type = "b",
  col = c("blue", "red"),
  pch = c(19, 17),
  trace.label = "Typ UI",
  xlab = "Latencia siete",
  ylab = "Priemerný čas dokončenia (sek.)",
  main = "Vplyv latencie a UI na čas dokončenia úlohy",
)
```




```
ggline(
  anova2,
  x = "latencia",
  y = "cas",
  group = "ui",
  color = "ui",
  palette = c("#E41A1C", "#377EB8"),
  plot_type = "b",
  xlab = "Latencia siete",
  ylab = "Priemerný čas dokončenia (sek.)",
  main = "Vplyv latencie a UI na čas dokončenia úlohy"
) +
  theme_bw()
```


Vplyv latencie a UI na cas dokoncenia úlohy
Priemerný cas dokoncenia (sek.)
50
40
30
20
Nízka
Vysoká
Latencia siete
ui
Jednoduché
Zložité